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CS 621 DATA ANALYTICS

Time Series Data

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TIME SERIES DATA/EVENTS

- A sequence of numbers that were associated with different time points, e.g., stock prices, temps.
- Date 2023-08-12
- Time 07:43:27
- Datetime 2023-08-12 07:43:27
- Unix Timestamp 1691858822

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STOCK PRICE DATA

	AAPL	MSFT	XOM	SPX
1/2/2003 0:00	7.4	21.11	29.22	909.03
1/3/2003 0:00	7.45	21.14	29.24	908.59
1/6/2003 0:00	7.45	21.52	29.96	929.01
1/7/2003 0:00	7.43	21.93	28.95	922.93
1/8/2003 0:00	7.28	21.31	28.83	909.93
1/9/2003 0:00	7.34	21.93	29.44	927.57
1/10/2003 0:00	7.36	21.97	29.03	927.57
1/13/2003 0:00	7.32	22.16	28.91	926.26
1/14/2003 0:00	7.3	22.39	29.17	931.66

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DATETIME MODULE

- The datetime module has the following useful classes to manage date and time.
- from datetime import date
- from datetime import time
- from datetime import datetime
- from datetime import timedelta

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DATE OBJECT

- Create a date object:
  - date(2001, 8, 11)
- Get today's date:
  - today = date.today()
- Get the current year:
  - today.year
- Convert a date to a string:
  - today.strftime('%Y-%m-%d')
- Present the weekday as an integer:
  - today.isoweekday()
- Show the name of the month:
  - today.strftime('%B')

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TIME OBJECT

- Create a time object:
  - now = time(9, 11, 4)
- Get the current time:
  - datetime.now().time()
- Get the hour:
  - now.hour
- Convert a time object to a string:
  - now.isoformat())

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DATETIME OBJECT

- Create a datetime object:
  - datetime(2023, 8, 11, 8, 45, 32)
- Get the current datetime:
  - now = datetime.now()
- Get the current month:
  - now.month
- Convert a datetime object to a string:
  - now.strftime('%H:%M:%S %Y-%m-%d')
- Get the timestamp of now:
  - now.timestamp()
- Generate a datetime object by giving a timestamp value:
  - datetime.fromtimestamp(1584689468)

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TIMEDELTA OBJECT

- Create datetime objects:
  - today = datetime(2023, 8, 11), yesterday = datetime(2023, 8, 10), tomorrow = datetime(2023, 8, 12)
- Compare dates like:
  - today > yesterday
  - today < tomorrow
  - yesterday == tomorrow
- Create a timedelta object:
  - dif = today – yesterday
- Add/Subtract a time difference using timedelta
  - today + timedelta (days = 2)
- Timeselta duration in seconds
  - diff.total_seconds()

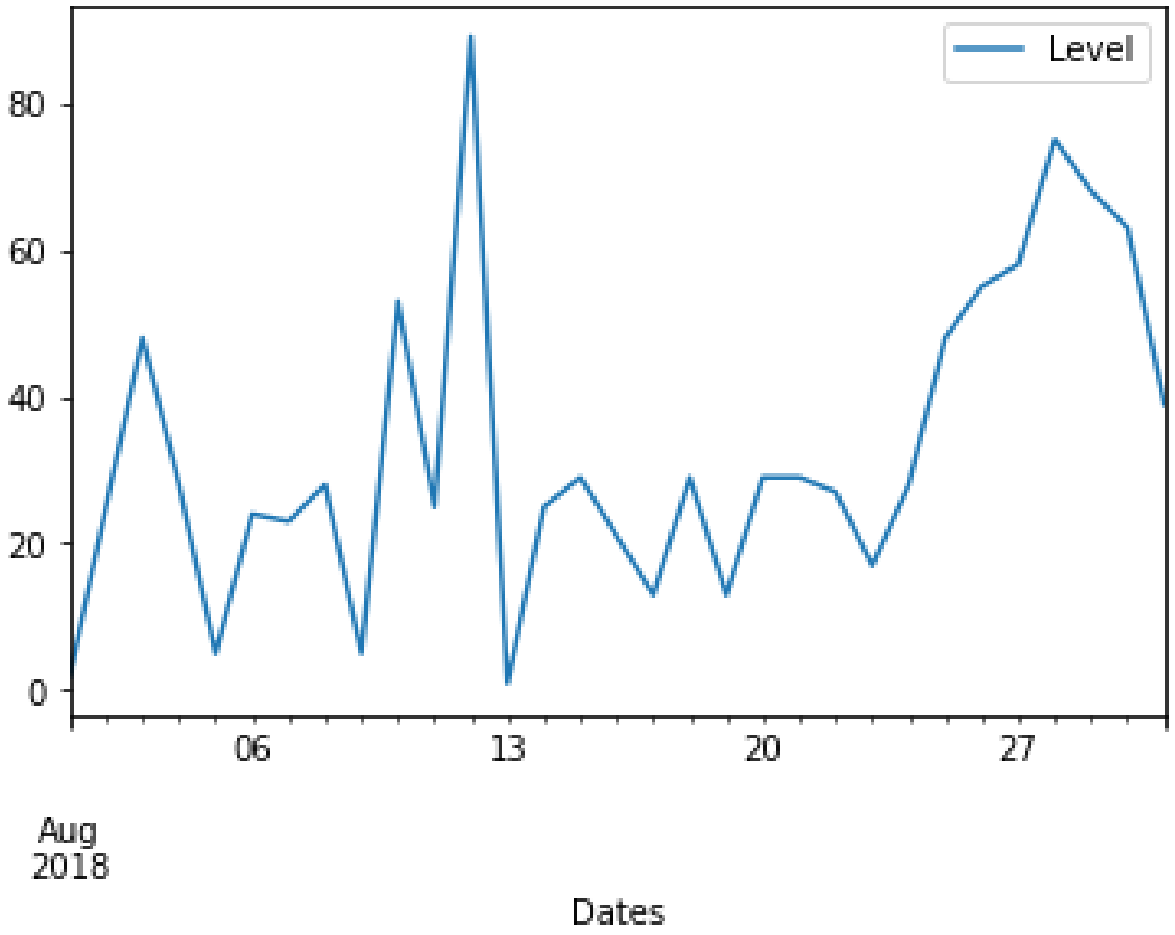
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TIME SERIES DATA EXAMPLE 2

Index	Date	Level
0	2018-08-01	2
1	2018-08-02	25
2	2018-08-03	48
3	2018-08-04	28
4	2018-08-05	5
5	2018-08-06	24
6	2018-08-07	23
7	2018-08-08	28
8	2018-08-09	5
9	2018-08-10	53

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TIME SERIES DATA VISUALIZATION



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ROLLING/MOVING/SLIDING WINDOWS

- Windowing technique allows us to spot patterns in time series data.
- Choose appropriate window size.
- Where to place the window? Or where to start the window?
- Compute the following within the window:
  - mean
  - std
  - max
  - min

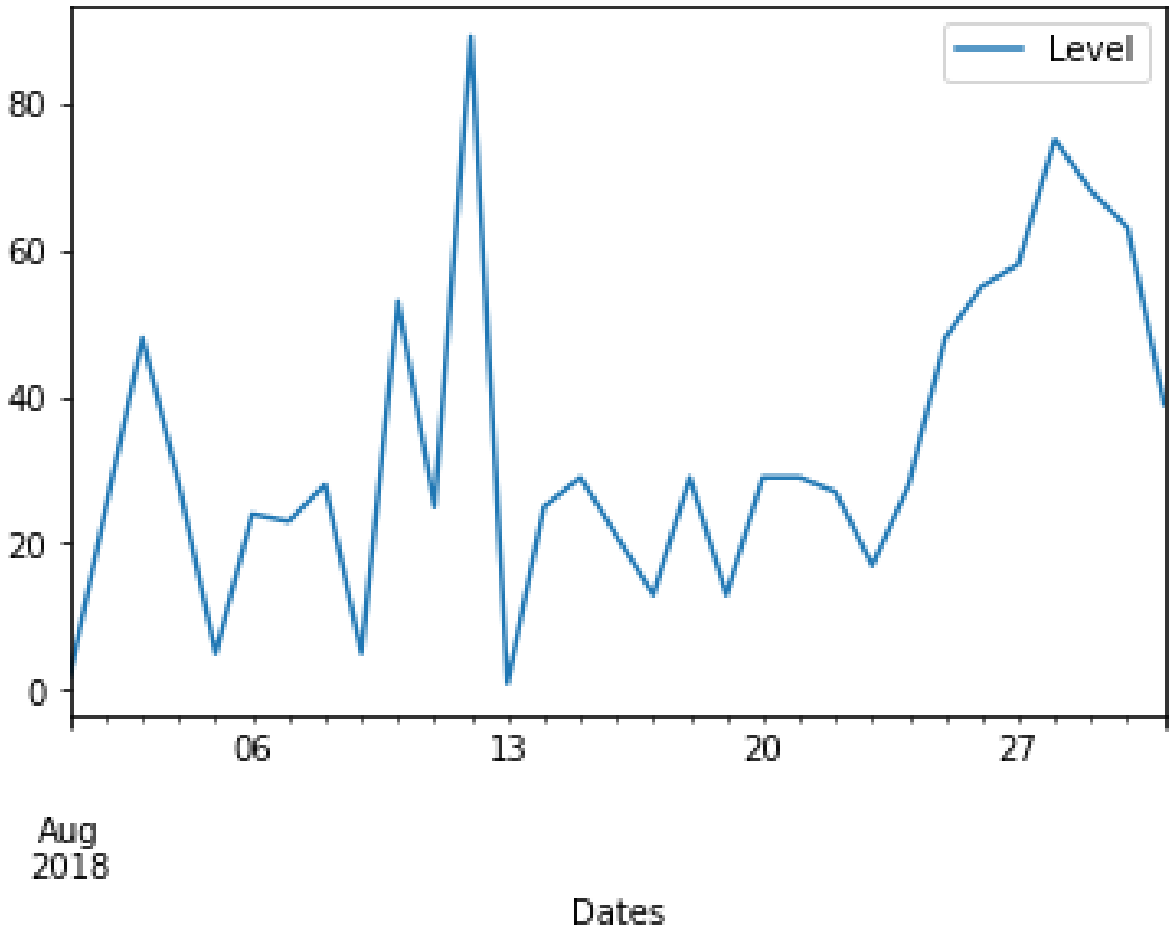
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TIME SERIES DATA IN PANDAS

- Load timeseries data to a dataframe. For example:
  - df = pd.read_csv('stock_px.csv', parse_dates=True)
- Observe trend via sliding window techniques.
- df.rolling(window=?).mean()
- df.rolling(window=?).max()
- df.rolling(window=?).min()
- df.rolling(window=?).std()

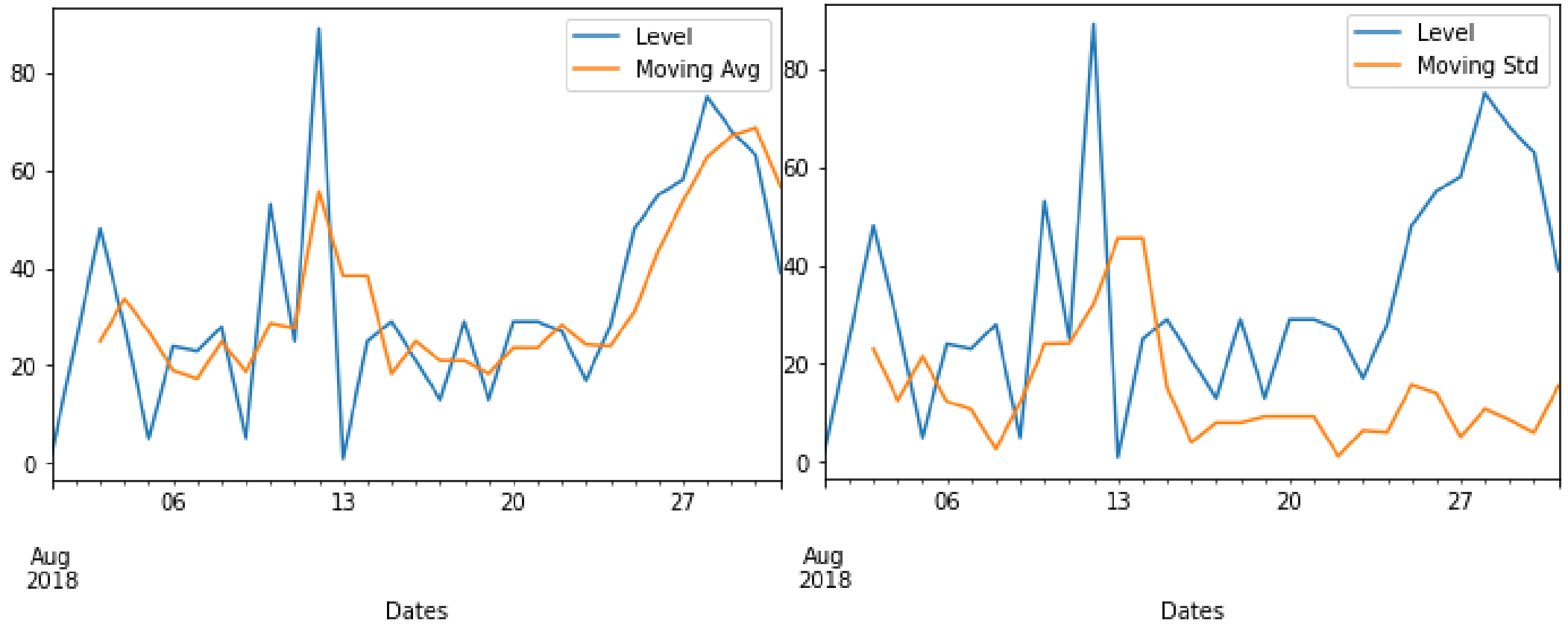
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TIME SERIES DATA VISUALIZATION



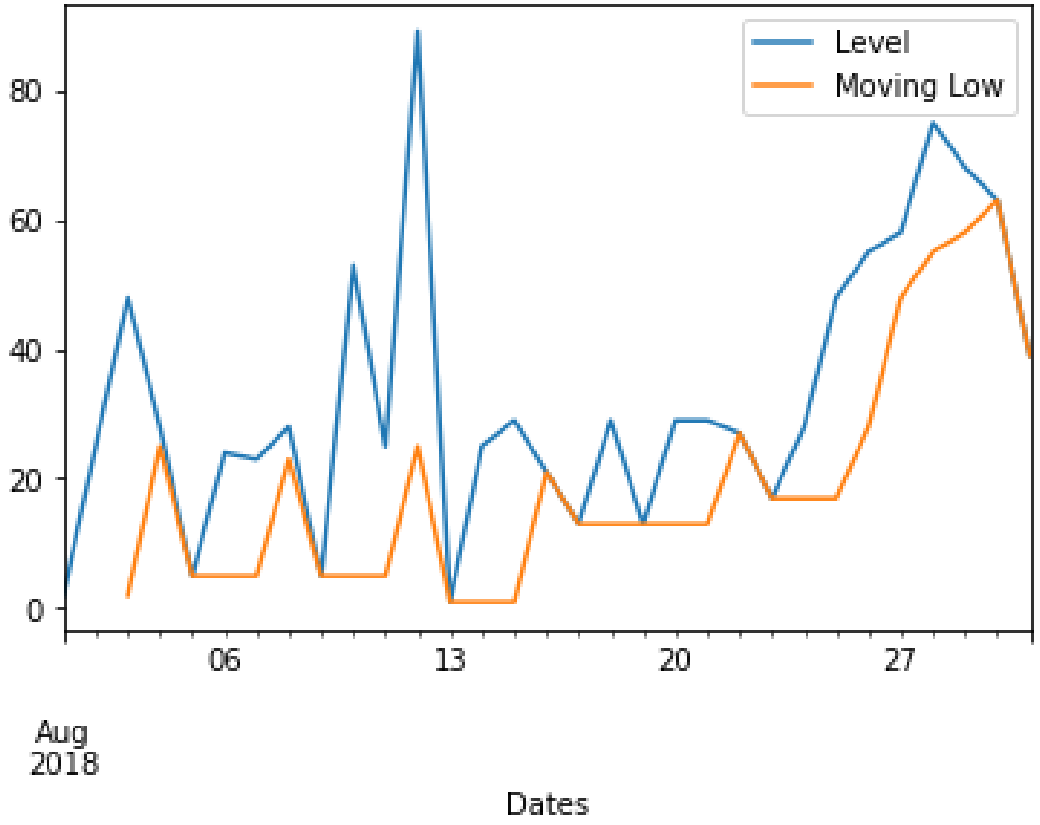
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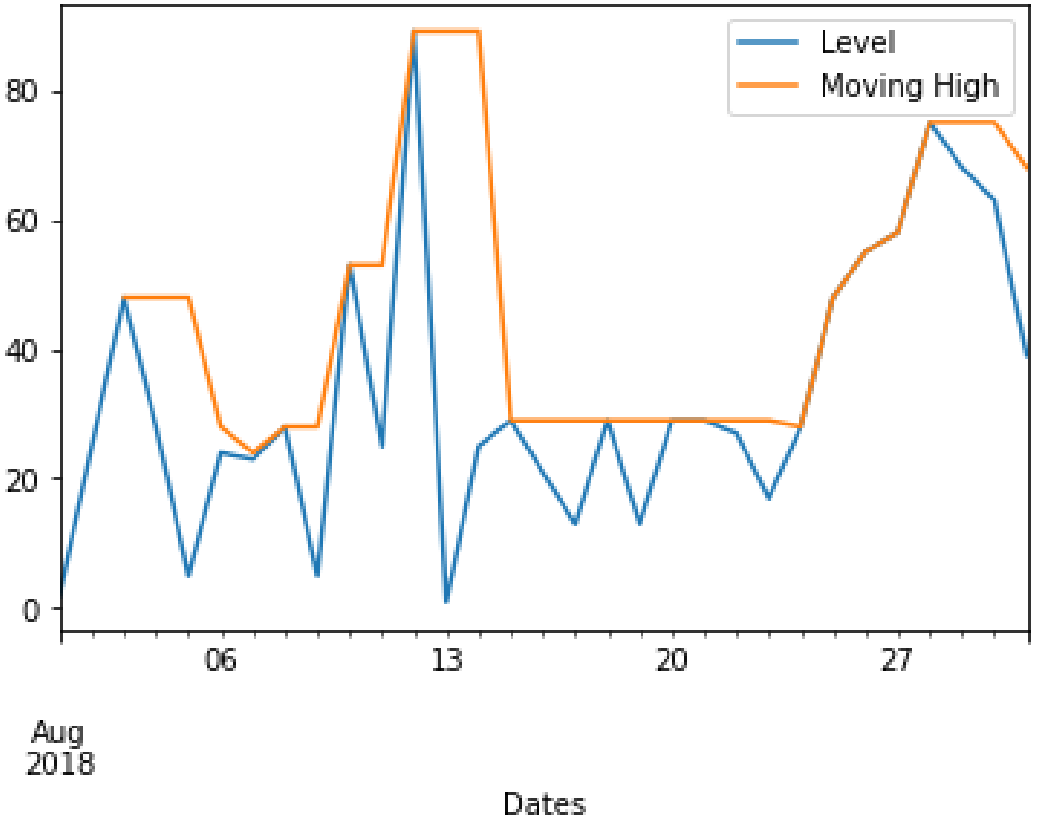
TIME SERIES DATA VISUALIZATION VIA ROLLING WINDOW TECHNIQUES



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TIME SERIES DATA VISUALIZATION VIA ROLLING WINDOW TECHNIQUES







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